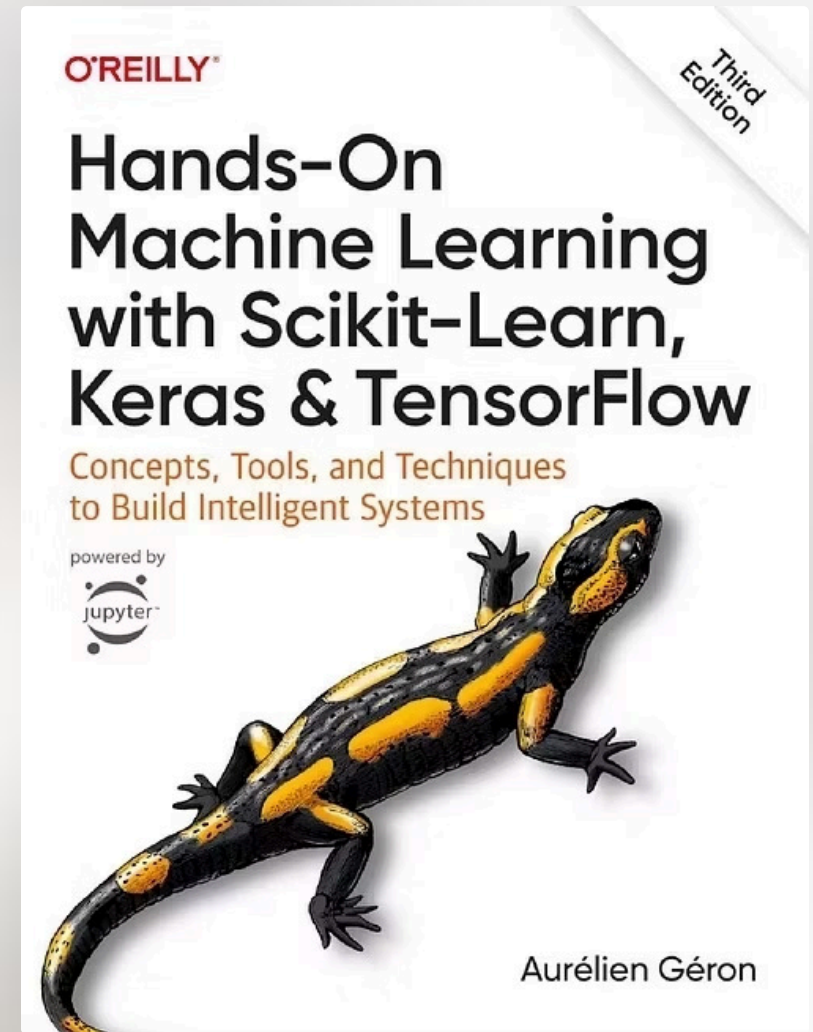


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/07/2025



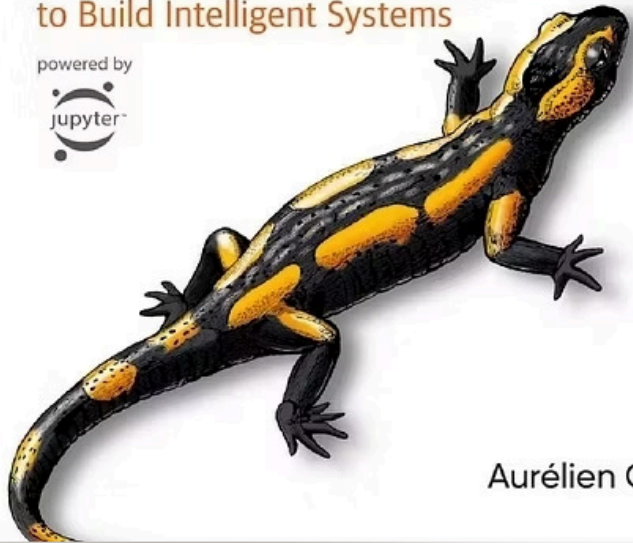
O'REILLY®

Third
Edition

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

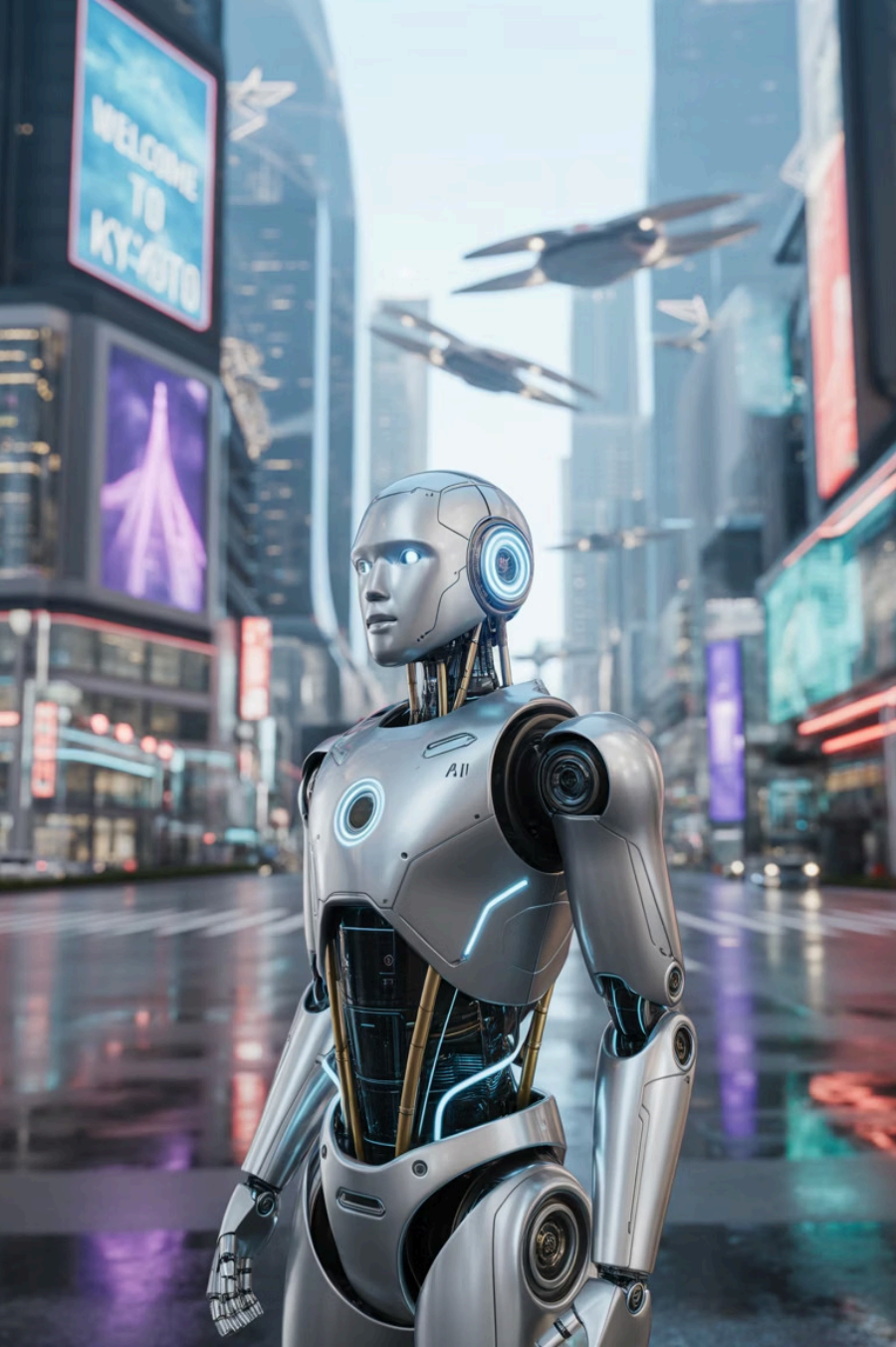
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Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape



Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
- 4. Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

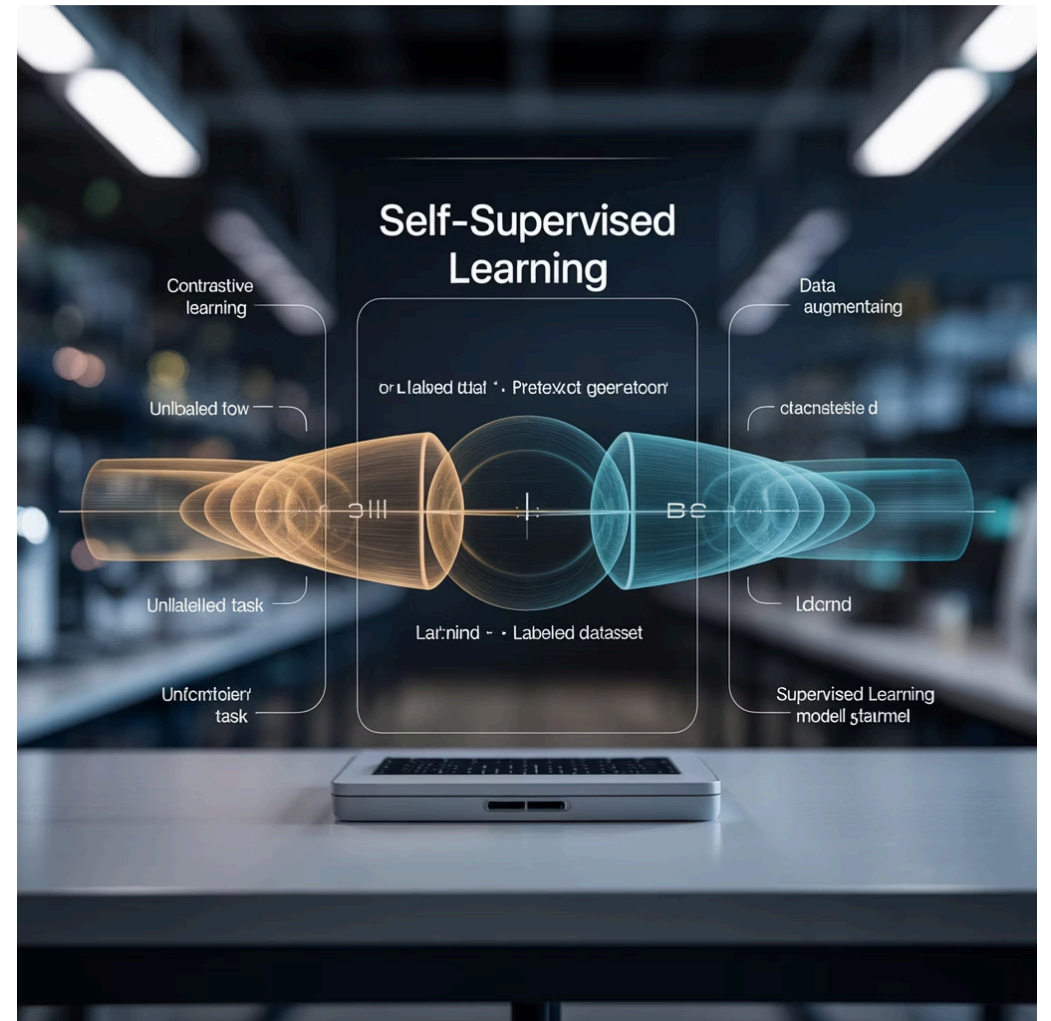
1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - **Classification criteria #1:** Supervision during training
 - **Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning**
 - **Classification criteria #2:** Approach to Learning Method
 - Batch and Online Learning
 - **Classification criteria #3:** Learning Method
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Classification criteria #1: Supervision during training

Supervised, Unsupervised, Semi-Supervised, **Self-Supervised**, and Reinforcement Learning

Self-Supervised Learning: Overview



Definition

- Generates a fully labeled dataset from an fully unlabeled one
- Can be used to train supervised learning algorithms once the dataset is labeled

Goal

- Create useful models from unlabeled data for subsequent fine-tuning on a task of interest

Example: Image Recovery with Self-Supervised Learning

- **Process:**
 - Mask a small part of each image in a large unlabeled dataset
 - Train a model to recover the original, unmasked part of the image
- **Inputs and Labels:**
 - Masked images are used as inputs
 - Original, unmasked images serve as labels



Self-supervised Learning

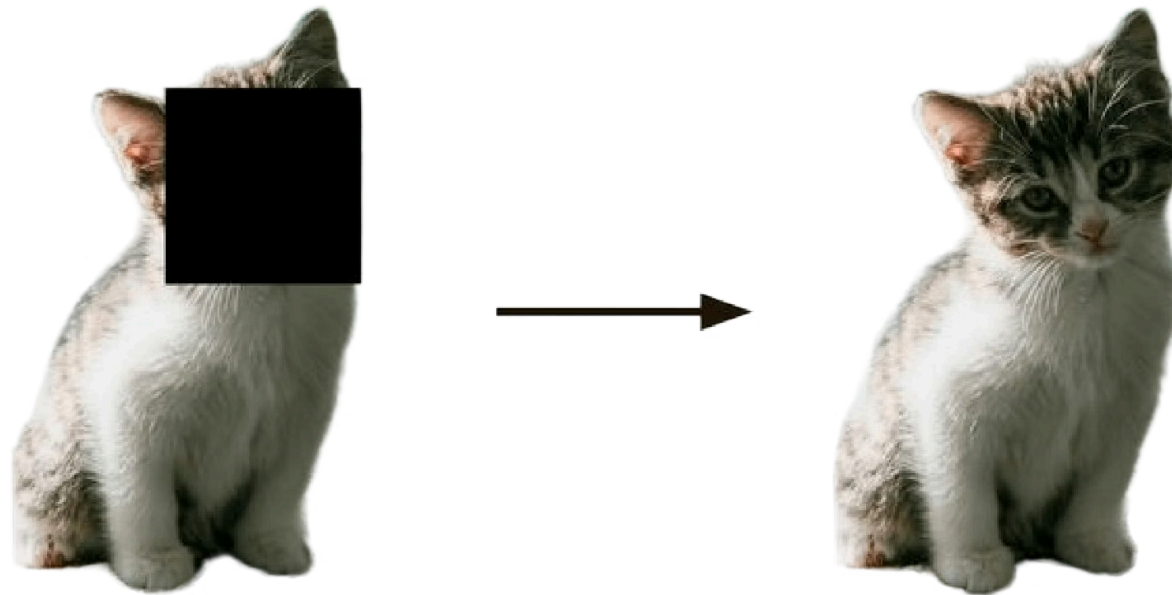


Figure 1-12. Self-supervised learning example: input (left) and target (right)



Applications of Self-Supervised Models

- **Direct Applications:**
 - Repairing damaged images
 - Erasing unwanted objects from images
- **Fine-Tuning Models for Other Tasks:**
 - **Example:** Fine-tuning the model for **object detection** or **classification tasks**



Example: Self-Supervised Learning for Pet Classification

Self-Supervised Learning for Pet Classification

Initial Task:

- **Goal:** Train a model to classify pets into species (e.g., cat, dog).
- **Approach:** Start with an **image-repairing** model using **self-supervised learning** on unlabeled pet photos.

Training Process:

- The model learns to repair images (e.g., by predicting missing parts, such as a cat's face).
- As the model repairs images, it **learns to distinguish between different species**.



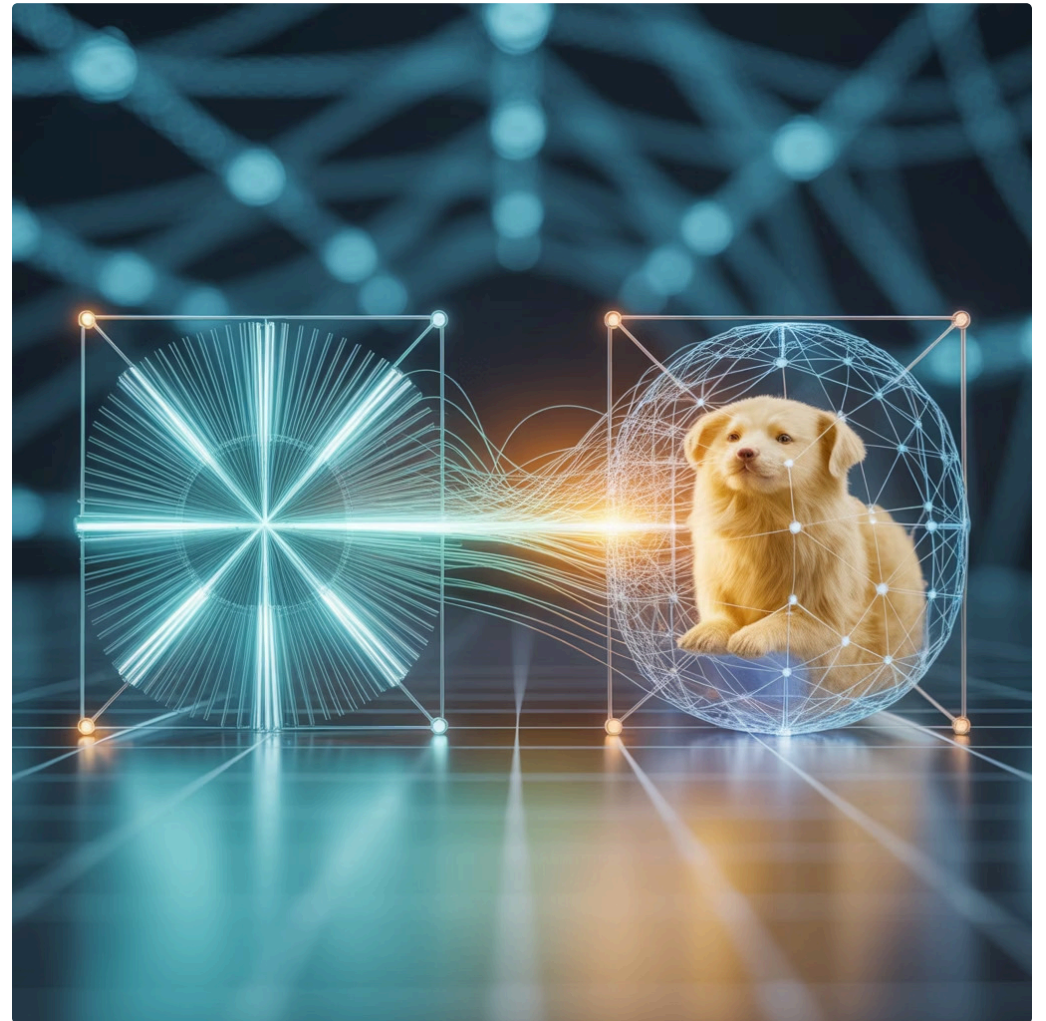
Transfer Learning

Definition:

- **Transfer Learning** involves **transferring knowledge** from one **task** to **another**.
- Once the **image-repairing model** is trained, it can be **fine-tuned** to **predict pet species** instead of repairing images.

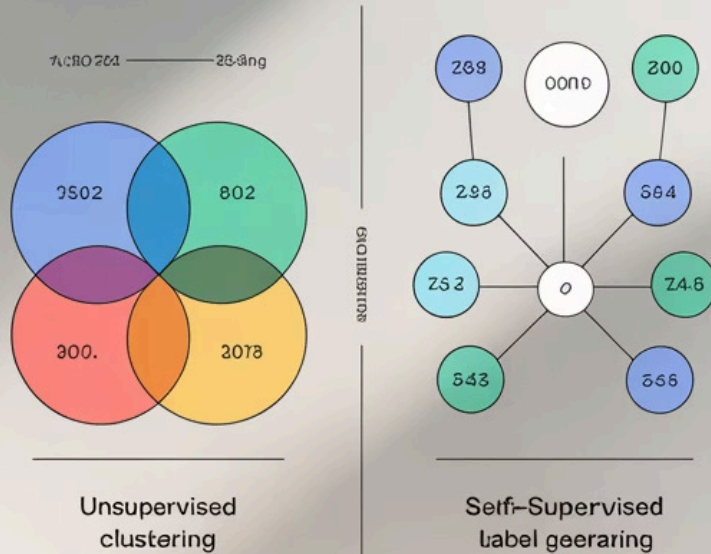
Final Step:

- Fine-tune the model on a **labeled dataset** to learn the mapping between the pet species it recognizes (e.g., cat, dog).



Self-Supervised vs. Unsupervised Learning

Self-supervised vs. Unsupervised learning



Understanding Machine Learning Approaches

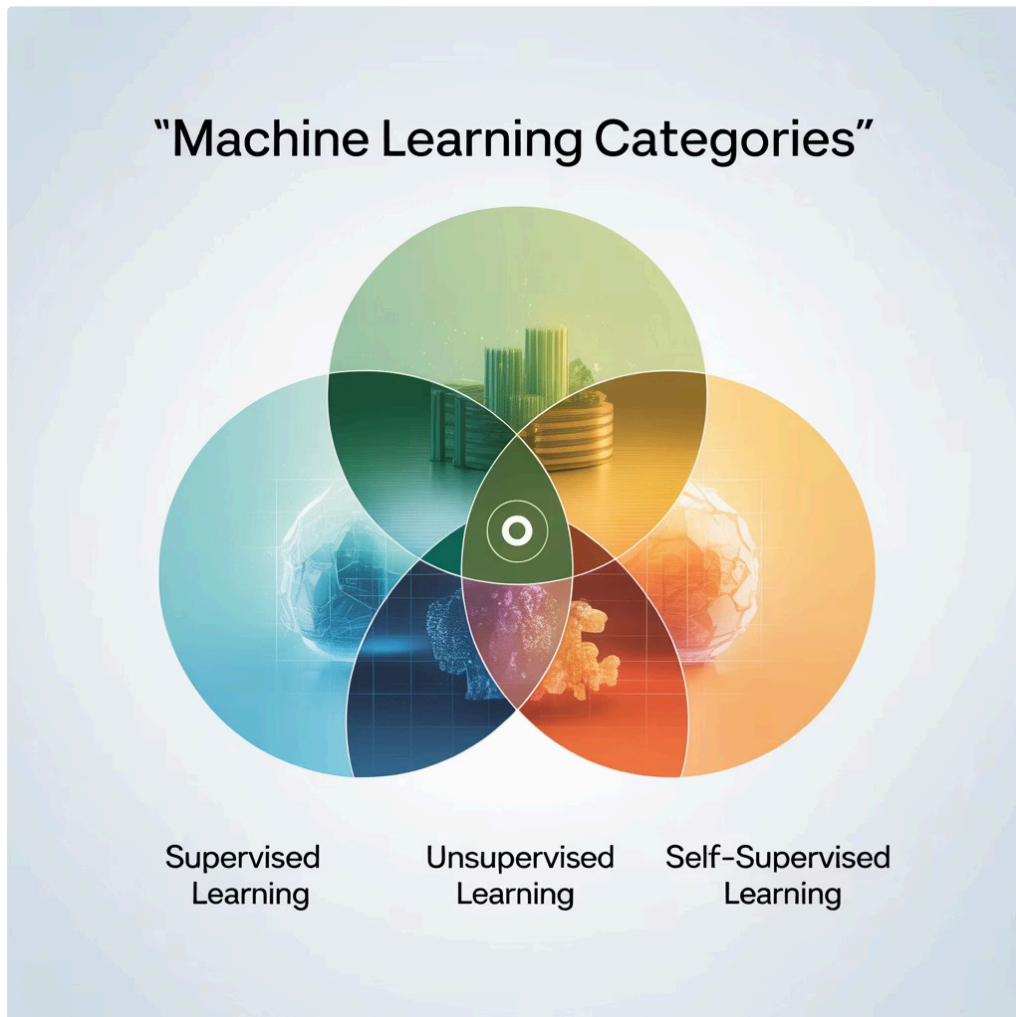
Self-Supervised Learning:

- Deals with fully **unlabeled** datasets, but generates **labels** during training.
- Primarily focuses on **tasks** similar to **supervised learning: classification** and **regression**.

Unsupervised Learning:

- Used for tasks like **clustering**, **dimensionality reduction**, and **anomaly detection**.
- Does not use labels for training.

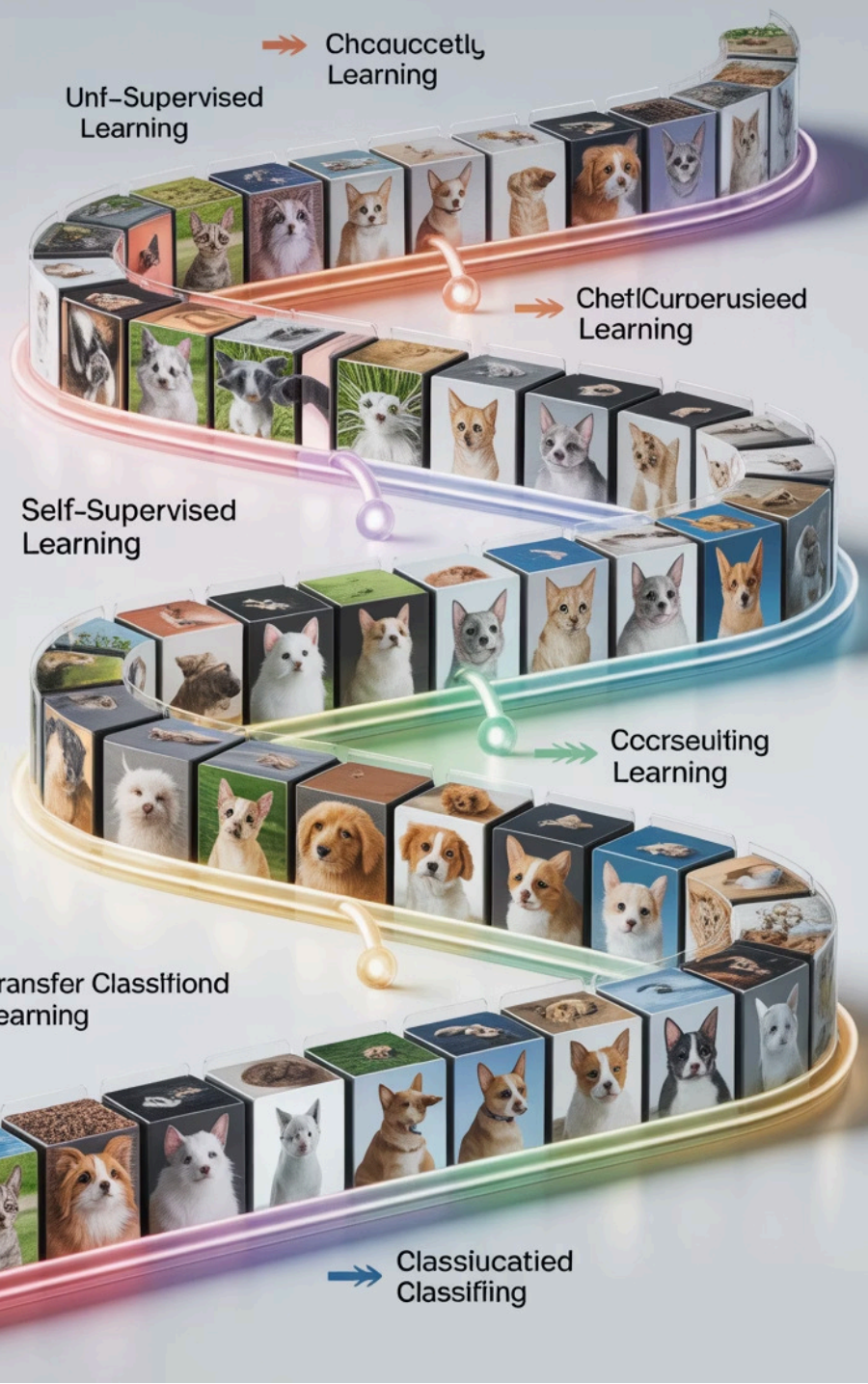
Self-Supervised Learning: A Separate Category



Why It's Different:

While it deals with **unlabeled data**, self-supervised learning uses **generated labels** during training.

Best treated as its own category, distinct from both **supervised** and **unsupervised learning**.



Self-Supervised Learning: Key Takeaways



Self-Supervised Learning:

Can be used to train models on **unlabeled data**, with the option to fine-tune them for specific **tasks** later.



Transfer Learning:

Key technique to **transfer knowledge** from **one task** to **another**, especially when using **deep neural networks**.

Self-Supervised Learning: Key Takeaways (cont.)



Data Generation:

- Transforms unlabeled data into a labeled dataset for supervised learning



Training Process:

- Involves training models on artificially labeled data (e.g., masked images)



Final Goal:

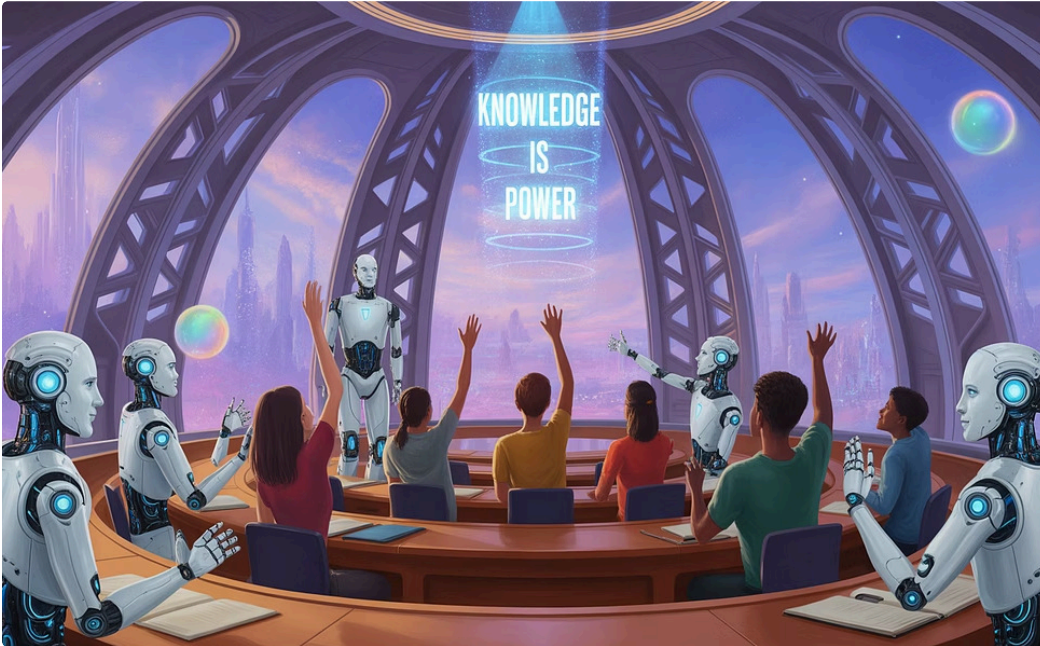
- Typically, the model is fine-tuned for a different, more specific task





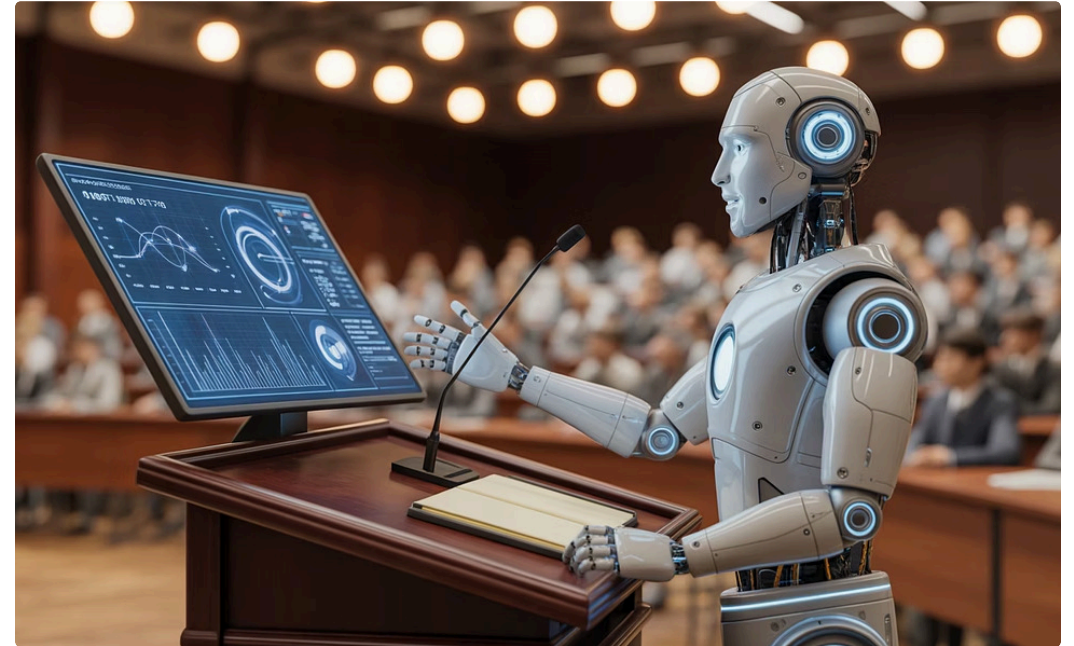
Next: Batch and Online Learning

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts